

Channel relaxation and stationary-state spectral gaps

Internal discussion note

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(Dated: October 5, 2026)

We distinguish the relaxation gap of the trajectory-averaged quantum channel from the occupation gap of its stationary correlation matrix. For the fixed-order, perfect-reset circuit, the charge-neutral and parity-even channel gaps follow from a product of single-particle projectors. The stationary occupation spectrum is a different object: a mode at occupation $1/2$ need not be a slow dynamical mode. We give an explicit channel-to-Lindbladian construction that preserves stationary states and controls the relaxation gap, explain what continuity alone does not establish, and collect the relevant literature with its assumptions. A matched hard-wall calculation illustrates the distinction without asserting a thermodynamic purity-gap closing from finite-size data.

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I. WHICH SPECTRUM IS GAPPED?

Throughout, one time step is one complete measurement cycle. After discarding the measurement outcomes and tracing out fresh ancillary modes, let $\tilde{\mathcal{E}}$ be the physical-layer channel. We first fix an invariant operator sector and assume relaxation to a unique stationary density matrix in that sector. If μ_ν are its nonstationary eigenvalues, define

$$r_{\text{MB}} = \max_{\nu \text{ nonstationary}} |\mu_\nu|, \quad g_{\text{MB}} = 1 - r_{\text{MB}}, \quad (1)$$

$$\kappa_{\text{MB}} = -\log r_{\text{MB}}, \quad g_{\text{MB}} = 1 - e^{-\kappa_{\text{MB}}}. \quad (2)$$

The channel gap g_{MB} is dimensionless; κ_{MB} is an asymptotic decay rate per cycle. Earlier project plots labeled

the corresponding two-point rate Δ_C ; here we write κ_C to keep it distinct from the correlation-matrix gap Δ . For complex multipliers, the relaxation gap is the deficit of the largest nonstationary *modulus* from unity, not $\min_\nu |1 - \mu_\nu|$. A multiplier near -1 can relax slowly despite being far from 1 in the complex plane. Additional unit-modulus modes obstruct mixing; they cannot be discarded to manufacture a positive gap.

The stationary correlation matrix is instead

$$(\overline{G}_{\text{ss}})_{ij} = \text{Tr}(\hat{\rho}_{\text{ss}} \hat{\psi}_i^\dagger \hat{\psi}_j), \quad \text{spec}(\overline{G}_{\text{ss}}) = \{n_a\} \subset [0, 1]. \quad (3)$$

We use the following explicit conventions:

$$\Delta = \min_a |n_a - \tfrac{1}{2}|, \quad \Delta_{\text{pur}} = \min_a |1 - 2n_a| = 2\Delta. \quad (4)$$

Thus Δ_{pur} is a *two-point purity gap*, not the purity $\text{Tr} \hat{\rho}^2$. For a Gaussian state, the single-particle modular energies are $\epsilon_a = \log[(1 - n_a)/n_a]$, so a crossing of $1/2$ is also a zero of the Gaussian modular spectrum. For a non-Gaussian state, this identifies the modular spectrum of the Gaussian state with the same two-point function, not that of the actual many-body density matrix. Covariance closure alone does not establish Gaussianity.

Neither n_a nor ϵ_a is a channel eigenvalue. In particular, an occupation- $1/2$ mode is maximally mixed at the two-point level; it need not be conserved or dark. A pure Gaussian trajectory has a projector-valued *full-system* G , whereas its subsystem correlation matrix may have occupations near $1/2$. Averaging trajectories and restricting a pure state are different operations. An occupation crossing alone also does not prove algebraic spatial correlations or a dissipative critical point.

II. EXACT RESET CIRCUIT AND ITS TWO-POINT MAP

We retain the overcomplete Wannier (OW) construction and physical-layer conventions of Bhuiyan, Pan, and Jian [1]. Distinct OW modes need not be orthogonal, and their number stabilizers need not commute. Let

$\{\hat{\psi}_i, \hat{\psi}_j^\dagger\} = \delta_{ij}$ and

$$\hat{\chi}_j = \sum_i v_{ji} \hat{\psi}_i, \quad v_j^\dagger v_j = 1, \quad \hat{N}_j = \hat{\chi}_j^\dagger \hat{\chi}_j. \quad (5)$$

Perfect measurement and feedback reset this mode to $s = 0$ or $s = 1$:

$$\begin{aligned} \tilde{\mathcal{E}}_{j,0}(\hat{\rho}) &= \hat{\chi}_j \hat{\rho} \hat{\chi}_j^\dagger + (\hat{\mathbf{1}} - \hat{N}_j) \hat{\rho} (\hat{\mathbf{1}} - \hat{N}_j), \\ \tilde{\mathcal{E}}_{j,1}(\hat{\rho}) &= \hat{\chi}_j^\dagger \hat{\rho} \hat{\chi}_j + \hat{N}_j \hat{\rho} \hat{N}_j. \end{aligned} \quad (6)$$

With $\mathcal{D}[\hat{L}](\hat{\rho}) = \hat{L} \hat{\rho} \hat{L}^\dagger - \frac{1}{2} \{\hat{L}^\dagger \hat{L}, \hat{\rho}\}$, the identities $\hat{N}_j^2 = \hat{N}_j$ and $\hat{\chi}_j \hat{\chi}_j^\dagger = \hat{\mathbf{1}} - \hat{N}_j$ give, exactly,

$$\begin{aligned} \tilde{\mathcal{E}}_{j,0} &= \text{Id} + \mathcal{D}[\hat{\chi}_j] + \mathcal{D}[\hat{N}_j], \\ \tilde{\mathcal{E}}_{j,1} &= \text{Id} + \mathcal{D}[\hat{\chi}_j^\dagger] + \mathcal{D}[\hat{N}_j]. \end{aligned} \quad (7)$$

This is not a first-order approximation to an exponential. The number dissipator is the measurement dephasing already present in the reset.

For $P_j = v_j v_j^\dagger$ and $Q_j = \mathbf{1} - P_j$, the adjoint dissipators acting on $\hat{\psi}_a^\dagger \hat{\psi}_b$ give

$$\begin{aligned} \delta_{\text{gain/loss}} \bar{G} &= -\frac{1}{2} (P_j \bar{G} + \bar{G} P_j) + s P_j, \\ \delta_{\text{dephas.}} \bar{G} &= P_j \bar{G} P_j - \frac{1}{2} (P_j \bar{G} + \bar{G} P_j). \end{aligned} \quad (8)$$

Here δ denotes the contribution of the specified dissipator, not a finite-time integration. Adding them to $\bar{G}$ yields

$$\bar{G}' = Q_j \bar{G} Q_j + s P_j. \quad (9)$$

Equivalently, $\tilde{\mathcal{E}}_{j,s} - \text{Id}$ generates $\dot{\bar{G}} = -P_j \bar{G} - \bar{G} P_j + P_j \bar{G} P_j + s P_j$. No sampling or Wick factorization is needed for this two-point equation.

The ordered product means composition, with step 1 acting first:

$$\tilde{\mathcal{E}} = \prod_{j=M}^1 \tilde{\mathcal{E}}_{j,s_j} \equiv \tilde{\mathcal{E}}_{M,s_M} \circ \dots \circ \tilde{\mathcal{E}}_{1,s_1}. \quad (10)$$

Define $A = Q_M \dots Q_1$ and $R_j = Q_M \dots Q_{j+1}$, with $R_M = \mathbf{1}$. Then

$$\bar{G}' = A \bar{G} A^\dagger + B, \quad B = \sum_j s_j R_j P_j R_j^\dagger. \quad (11)$$

The steady-state equation is affine, $\bar{G}_{\text{ss}} = A \bar{G}_{\text{ss}} A^\dagger + B$. Consequently, a stationary many-body state does *not* require an eigenvalue 1 of A . The eigenvalue 1 belongs to $\tilde{\mathcal{E}}$, not necessarily to A .

For deviations from stationarity, column vectorization gives

$$\text{vec}(\delta G') = (A^* \otimes A) \text{vec}(\delta G). \quad (12)$$

If a_p are the eigenvalues of A , the covariance multipliers are $a_p a_q^*$. Thus, with $r = \rho(A) = \max_p |a_p|$,

$$\boxed{g_C = 1 - r^2, \quad \kappa_C = -2 \log r.} \quad (13)$$

The square comes from the two-sided action on G . No eigenvalue of A is removed in computing r ; an actual unit-modulus mode gives $g_C = 0$. For $Aw = aw$, the matrix $ww^\dagger$ is a covariance eigenmode with multiplier $|a|^2$. It is not, in general, an eigenprojector of $\bar{G}_{\text{ss}}$. Nonnormality further distinguishes right state perturbations from left, observable relaxation modes.

III. WHEN IS THIS THE MANY-BODY GAP?

Covariance closure preserves the Heisenberg-space span of $\hat{\mathbf{1}}$ and $\hat{\psi}_i^\dagger \hat{\psi}_j$. Its spectrum is therefore included in the many-body spectrum. In a mixing sector containing these operators,

$$r_{\text{MB}} \geq r^2, \quad g_{\text{MB}} \leq g_C, \quad \kappa_{\text{MB}} \leq \kappa_C. \quad (14)$$

A finite upper bound alone does not establish a finite many-body gap: higher correlations could relax more slowly. This is the general spectral-inclusion statement, closely related to the closed correlation hierarchy for quadratic Lindbladians [2].

For the *particular exact resets* in Eq. (6), a stronger argument applies. In a basis containing the measured mode, a reset erases a single measured leg in an even operator; an occupation pair is replaced by s . A normal-ordered $2p$ -point neutral correlation therefore maps to itself and lower orders. The homogeneous block of one cycle is

$$H_p = (\wedge^p A) \otimes (\wedge^p A)^*, \quad \Lambda_{I,J} = \prod_{i \in I} a_i \prod_{j \in J} a_j^*, \quad (15)$$

up to vectorization order, with $|I| = |J| = p$. The two index sets may overlap. Antisymmetry produces the minors of A ; Cauchy–Binet composes the minors in the actual reset order. The hierarchy is block triangular, so lower-order source terms do not alter these eigenvalues. This statement also follows by Schur triangularization when A is defective.

The $p = 0$ block is the constant. If $r < 1$, every $p \geq 1$ multiplier satisfies $|\Lambda_{I,J}| \leq r^{2p} \leq r^2$, and $p = 1$ attains r^2 . These moments span the charge-neutral operator sector. More generally, an even monomial has at least two legs and obeys the same bound. Hence

$$\boxed{g_{q=0} = g_{\text{even}} = g_C.} \quad (16)$$

Charge neutrality means $[\hat{N}_F, \hat{\rho}] = 0$ and permits mixtures of number sectors; it is not strong particle-number conservation under gain and loss. Physical density matrices are parity even. For the specified Kraus extension to unrestricted operator space, the odd-sector spectrum instead reaches modulus r , giving $g_{\text{full}} = 1 - r$. The parity-similarity argument and full sector accounting are recorded in the companion [proof note](#).

Equation (16) requires perfect resets and the fixed repeated schedule. It is not a consequence of two-point

closure alone, and must not be exported without proof to imperfect feedback, additional dephasing, or randomly averaged schedules. A spectral gap also does not fix a size-uniform mixing time: nonnormal transients, Jordan factors, and size-dependent prefactors can matter. Nor does it determine the purification time of individual conditioned trajectories.

IV. NO UNIVERSAL IMPLICATION FROM THE STATIONARY SPECTRUM

A. Global depolarization: an exactly solvable channel

An elementary counterexample separates the two questions. For any density matrix $\hat{\sigma}$ and $0 < p \leq 1$, the completely positive trace-preserving map

$$\mathcal{R}_{p,\sigma}(\hat{X}) = (1-p)\hat{X} + p\text{Tr}(\hat{X})\hat{\sigma} \quad (17)$$

has unique stationary state $\hat{\sigma}$. Every traceless operator has multiplier $1-p$, so its channel gap is p , independent of the spectrum of $\hat{\sigma}$. Choosing $\hat{\sigma}$ maximally mixed gives $\Delta_{\text{pur}} = 0$ at a strictly positive dynamical gap. Conversely, $p \rightarrow 0$ closes the channel gap without changing the stationary state. For $\hat{\sigma} = \hat{\mathbf{1}}/d$, define $\mathcal{T}(\hat{X}) = \text{Tr}(\hat{X})\hat{\mathbf{1}}/d$. Since $\mathcal{T}^2 = \mathcal{T}$, the associated generator has the exact solution

$$\begin{aligned} \mathcal{L} &= \gamma(\mathcal{T} - \text{Id}), \\ e^{t\mathcal{L}} &= \mathcal{T} + e^{-\gamma t}(\text{Id} - \mathcal{T}). \end{aligned} \quad (18)$$

Its eigenvalues are 0 once and $-\gamma$ on every traceless operator. The Lindbladian gap is γ , and its time- τ channel gap is $1 - e^{-\gamma\tau}$, while the unique endpoint is maximally mixed. For N fermionic modes, $d = 2^N$ and $\bar{G}_{\text{ss}} = \mathbf{1}/2$, so $\Delta_{\text{pur}} = 0$ exactly. This global example is not a locality theorem.

B. On-site depolarization: a size-independent local example

For a lattice of qubits, let $\mathcal{T}_i$ replace site i by $\hat{\mathbf{1}}_i/2$ while leaving the reduced state of all other sites unchanged. The local generators commute, and

$$\mathcal{L}_{\text{loc}} = \gamma \sum_i (\mathcal{T}_i - \text{Id}) = \frac{\gamma}{4} \sum_{i,a=x,y,z} \mathcal{D}[\hat{\sigma}_i^a]. \quad (19)$$

A Pauli string with w nonidentity factors is an eigenoperator with eigenvalue $-\gamma w$. Thus the unique stationary state is $\hat{\mathbf{1}}/2^N$, the many-body Lindbladian gap is γ , and $e^{\tau\mathcal{L}_{\text{loc}}}$ has channel gap $1 - e^{-\gamma\tau}$, all independent of N . This is a genuine bounded-range counterexample, not a global reset in disguise. It is stated for a tensor-product spin lattice; no locality-preserving spin-fermion identification is assumed.

A directly fermionic one-mode version is balanced loss and gain,

$$\mathcal{L}_{\text{bal}} = \eta(\mathcal{D}[\hat{c}] + \mathcal{D}[\hat{c}^\dagger]), \quad \dot{n} = \eta(1 - 2n). \quad (20)$$

Its stationary state is $\hat{\mathbf{1}}/2$ and its full operator spectrum is $\{0, -\eta, -\eta, -2\eta\}$. The full gap is η , or 2η on the physical parity-even single-mode sector. In either convention it is strictly positive although $n_{\text{ss}} = 1/2$. This is the simplest gain/loss realization of the same distinction.

C. If “Hamiltonian gap” is meant literally

A gapped Hamiltonian also admits a maximally mixed *thermal* state: take $\hat{H} = \varepsilon \sum_i \hat{c}_i^\dagger \hat{c}_i$, with $\varepsilon > 0$. Its many-body excitation gap is ε , but at inverse temperature $\beta = 0$, $\hat{\rho}_\beta = \hat{\mathbf{1}}/2^N$. There is no contradiction: an excitation gap is not a purity gap of a finite-temperature state. Unitary evolution under $\hat{H}$ does *not* relax every initial state to that endpoint; it preserves the density-matrix eigenvalues. The first two examples, in contrast, really do have unique attractive maximally mixed endpoints.

D. Scope of the counterexamples

Within Gaussian channel theory, the same freedom is visible in $G' = AGA^\dagger + B$ [3]. For example, $A = \sqrt{q}\mathbf{1}$, $B = (1-q)G_0$, with $0 \leq q < 1$ and $0 \leq G_0 \leq \mathbf{1}$, is an admissible covariance map with fixed point G_0 and $g_C = 1 - q$. The source and contraction control different properties. This is an illustration within that channel class, not a replacement for the reset-sector proof above.

Locality, symmetry, purity, or detailed balance can impose additional relations. Such relations require their actual hypotheses; they do not follow from the word “gap.” In particular, results linking rapid local mixing to clustering [4] concern spatial correlations, not simply whether an eigenvalue of $\bar{G}_{\text{ss}}$ equals $1/2$.

V. A CONTROLLED CHANNEL-LINDBLADIAN DEFORMATION

There is a precise version of making each complete cycle probabilistic. Given a channel $\tilde{\mathcal{E}}$, define

$$\tilde{\mathcal{E}}_p = (1-p)\text{Id} + p\tilde{\mathcal{E}}, \quad 0 < p \leq 1. \quad (21)$$

Its fixed space is exactly that of $\tilde{\mathcal{E}}$, since $\tilde{\mathcal{E}}_p - \text{Id} = p(\tilde{\mathcal{E}} - \text{Id})$. Its eigenvalues are $\mu_\nu(p) = 1 - p + p\mu_\nu$. If the original mixing-sector gap is g , then

$$|\mu_\nu(p)| \leq 1 - p + p|\mu_\nu| \leq 1 - pg, \quad g(p) \geq pg. \quad (22)$$

Thus a gap already bounded below is preserved for $p > 0$. This is a quantitative statement, not an appeal to continuity.

Set $p = \gamma dt$, where γ is the attempted-cycle rate. The continuous-time limit is

$$\mathcal{L}_{\text{cycle}} = \gamma(\tilde{\mathcal{E}} - \text{Id}). \quad (23)$$

If $\tilde{\mathcal{E}}(\hat{\rho}) = \sum_a \hat{K}_a \hat{\rho} \hat{K}_a^\dagger$, completeness immediately gives

$$\mathcal{L}_{\text{cycle}} = \sum_a \mathcal{D}[\sqrt{\gamma} \hat{K}_a], \quad e^{t\mathcal{L}_{\text{cycle}}} = e^{-\gamma t} \sum_{n=0}^{\infty} \frac{(\gamma t)^n}{n!} \tilde{\mathcal{E}}^n. \quad (24)$$

This is a legitimate Lindblad generator [5]: complete cycles occur at Poisson-distributed times. Its stationary states are unchanged, and its eigenvalues are $\lambda_\nu = \gamma(\mu_\nu - 1)$. Therefore

$$\delta_{\mathcal{L}} = \gamma \min_{\nu \text{ nonstationary}} (1 - \text{Re } \mu_\nu) \geq \gamma g. \quad (25)$$

For our covariance map, r^2 is itself a positive real multiplier and all other moduli are at most r^2 . Consequently,

$$g_C(p) = p(1 - r^2), \quad \delta_{\mathcal{L},C} = \gamma(1 - r^2). \quad (26)$$

The same equalities hold in the neutral/even many-body sector under Eq. (16). The Poissonized rate is not $\gamma\kappa_C$: random cycle counts and deterministic cycle counts have different asymptotic rates.

The vanishing of $g(p)$ as $p \rightarrow 0$ merely reflects a vanishing chance of updating per discrete step. Measured in physical time with $p = \gamma dt$, the generator gap stays finite. At exactly $p = 0$, the discrete channel is the identity and its fixed space enlarges; that endpoint must not be called a mixing channel.

Conversely, if a relaxing Lindbladian has gap $\delta_{\mathcal{L}} > 0$, its finite-time channel $\tilde{\mathcal{E}}_\tau = e^{\tau\mathcal{L}}$ satisfies

$$g(\tau) = 1 - e^{-\tau\delta_{\mathcal{L}}}, \quad \tau > 0, \quad (27)$$

in the same sector. This does not mean an arbitrary prescribed channel is $e^{\tau\mathcal{L}}$ for some time-independent Lindbladian. Semigroup embeddability is a separate restriction [6]; the construction $\gamma(\tilde{\mathcal{E}} - \text{Id})$ does not assert that $\log \tilde{\mathcal{E}}$ is Lindbladian.

A. Two limitations that matter here

First, Poissonizing a *whole raster cycle* need not produce a local sum of bounded-range generators. Its Kraus operators are full-cycle products, and the sequential depth grows with system size. Instead weakening each local reset gives

$$\begin{aligned} & \prod_{j=M}^1 [\text{Id} + \gamma_j dt (\tilde{\mathcal{E}}_j - \text{Id})] \\ &= \text{Id} + dt \sum_j \gamma_j (\tilde{\mathcal{E}}_j - \text{Id}) + O(dt^2). \end{aligned} \quad (28)$$

TABLE I. Dimensionless channel gap and full-system two-point purity gap at cycle 61. $N_x = 20$ is fixed. The displayed digits summarize deterministic calculations, not statistical error bars.

N_y	$\alpha_1 = 1$		$\alpha_1 = 3$	
	g_C	Δ_{pur}	g_C	Δ_{pur}
20	0.851086	0.086790	0.961228	0.985595
40	0.841480	0.004357	0.959010	0.985659
60	0.838258	0.021813	0.958243	0.985667
80	0.836644	0.005699	0.957852	0.985668
100	0.835675	0.009523	0.957611	0.985668

The resulting local sum generally has a different stationary state from the ordered finite-reset cycle. A common fixed state of every $\tilde{\mathcal{E}}_j$ would remove this particular obstruction, but must be demonstrated. The noncommuting OW stabilizers make that distinction important.

Second, continuity of a family of channels or generators alone cannot exclude gap closure along the path, or in the thermodynamic limit. To transfer a size-independent gap using Eq. (22), one needs $\inf_N g_N > 0$ and p bounded away from zero, or a fixed physical rate in the continuous-time limit. Neither a positive finite-size gap nor a smooth deformation proves this uniform bound. The construction therefore does not bypass locality-based obstructions to preparing pure topological states, nor does it establish a gap for a different local Lindblad approximation to our circuit.

VI. MATCHED HARD-WALL CALCULATION

We compare the exact ordered-product channel gap with the stationary two-point purity gap for $N_x = 20$, $N_y = 20, 40, 60, 80, 100$, and $\alpha_1 = 1, 3$. Both calculations use $\alpha_2 = 30$, $n_{\text{shell}} = 1$, inclusive walls $x = 5, 15$, hard-wall support truncation, all slabs active, periodic boundaries, zero twist, X trial orbitals, complex128, and perfect correction including measurement dephasing. The raster- y schedule uses x outermost, y innermost, and $A+, A-, B+, B-$ within each cell.

The canonical CPU method `classA_U1FGTN.run_markov_channel` evolves the initially maximally mixed correlation matrix for 60 cycles plus one comparison cycle. This is exact outcome-averaged two-point evolution, not a finite sample average. All ten cases pass the recorded stationarity checks: the last five absolute Frobenius increments are below 10^{-10} , and the purity gaps at cycles 60 and 61 differ by less than 2×10^{-10} . The largest final increment is 6.51×10^{-15} . These are numerical convergence checks, not a rigorous a posteriori bound on fixed-point error for a nonnormal map.

Table I shows a large relaxation gap for both alphas,

despite sharply different stationary occupation gaps. At $\alpha_1 = 1$, the latter is small and nonmonotonic over the tested sizes; at $\alpha_1 = 3$, it remains near 0.986. These data support a separation of the two spectral diagnostics. They do *not* yet prove that Δ_{pur} vanishes as $N_y \rightarrow \infty$, nor a nonzero limiting channel gap. Furthermore, fixed $N_x = 20$ is a fixed-width limit, not a two-dimensional thermodynamic limit.

The data also contain a separately postprocessed translation average along y . Its momentum-resolved occupations describe that twirled matrix, not automatically the actual raster-channel fixed point. Table I uses the untwirled full spectrum. Endpoint matrices, source hashes, and checksum-verified completion records are preserved in the matched campaign; the accompanying build record binds each row to those files. The existing [comparison plot](#) is kept separate from this note.

VII. ANNOTATED LITERATURE AND SCOPE

The following references address different parts of the argument. The linked titles and arXiv identifiers in the bibliography provide direct access; none should be cited as a theorem beyond its hypotheses.

a. Dissipative topology and the central counterexample. Bardyn *et al.*, Ref. [7], Sec. 3.3 and Sec. 7.2, especially Fig. 6, explicitly separate the purity and dissipative gaps. Their example has a bulk purity-gap closing while the bulk damping gap stays finite. This is the most direct precedent for our distinction. It concerns engineered Gaussian Lindblad dynamics, not the present ordered reset circuit. The qualifier “bulk” matters: open boundaries can support zero-damping Majorana modes, so the example must not be described as proving a gapped full open-boundary generator.

b. Damping, noise, and nonequilibrium criticality. Mittal *et al.*, Ref. [8], distinguish damping, noise, and purity scales [Eqs. (38)–(43)]. In the commuting Gaussian setting, the stationary covariance involves their ratio. Their special frustration-free/dark-state conditions lock noise and damping and give pure stationary states; they do not impose a general identity between purity and relaxation gaps. This is useful for explaining which extra structure can restore relations familiar from ground-state physics, rather than assuming such relations for arbitrary channels.

c. Symmetry of the generator versus symmetry of its state. Mao and Yang, Ref. [9], relate non-Hermitian symmetry classes of quadratic Lindblad dynamics to the Altland–Zirnbauer classes of unique Gaussian stationary states. Their correspondence concerns symmetry constraints, not equality of spectral gaps. The construction assumes quadratic Hamiltonians and linear gain/loss jumps. Our number-dephasing term is quadratic in fermions, and the cycle is ordered and discrete; two-point closure alone does not place it within that theorem.

d. Many-body spectra and closed correlation hierarchies. Barthel and Zhang, Ref. [2], Secs. III.5–III.6, distinguish unrestricted and two-point relaxation and develop a block-triangular hierarchy including Hermitian quadratic jumps (Proposition 7). That is the appropriate continuous-time reference for Eq. (14). The exact discrete-reset equality in Eq. (16) requires our additional higher-moment argument; it should not be attributed to their covariance equation alone.

e. Discrete fermionic channels. Dierckx, Fannes, and Pogorzelska, Ref. [3], Sec. 5.1, give the affine symbol maps and complete-positivity conditions for fermionic quasi-free channels. This is a direct channel, rather than only Lindbladian, reference for separating the contraction A from the source B . In the convention of Eq. (11), the gauge-invariant Gaussian-channel condition is $0 \leq B \leq 1 - AA^\dagger$. It does not by itself establish Gaussianity from a closed two-point equation.

f. Discrete channels with independent spectral diagnostics. Sá *et al.*, Ref. [10], study random Kraus maps and circuits. A transition in the support of the complex channel spectrum need not produce a corresponding stationary-state spectral transition. This is relevant channel-level evidence for separating the two diagnostics, but their annulus-to-disk transition is not necessarily a closing of the relaxation gap, nor a fermionic purity-gap crossing. Yoshimura and Sá, Ref. [11], study relaxation in noisy Floquet circuits, including depolarizing settings with a maximally mixed stationary state. Their order-of-limits discussion is useful when comparing weak-noise and thermodynamic limits; it is not a boundary occupation-gap theorem. Kukulski *et al.*, Ref. [12], provide broader context on random-channel spectra, gaps, and invariant states, rather than a universal identification of those spectra.

g. When a dynamical gap constrains spatial correlations. Kastoryano and Eisert, Ref. [4], prove clustering statements under local mixing hypotheses. Their spectral-gap result (Theorem 9, with a free-fermion corollary) uses regularity and reversibility assumptions; stronger mixing conditions control further correlation measures. This is not a theorem that any finite nonnormal-channel gap forces any chosen stationary occupation gap to remain open. Those locality and reversibility assumptions have not been established for the whole-cycle Poissonization here.

h. Continuous time and embeddability. Wolf’s notes, Ref. [5], provide the standard channel and Lindblad framework underlying Eq. (24). Puchała, Rudnicki, and Życzkowski, Ref. [6], discuss which Pauli channels belong to semigroups. They are useful for the warning that not every channel has a Lindbladian logarithm, not as the source of our elementary proof that $\gamma(\tilde{\mathcal{E}} - \text{Id})$ is a generator.

VIII. STATEMENT SUITABLE FOR THE MANUSCRIPT

The relaxation spectrum of the trajectory-averaged channel and the occupation spectrum of its stationary correlation matrix diagnose different properties. For the fixed-order perfect-reset circuit, the parity-even relaxation gap is $g_C = 1 - \rho(A)^2$, where A is the ordered product of the complementary OW projectors. The stationary matrix

instead solves an affine fixed-point equation containing the reset source. An occupation near $1/2$ therefore does not require a slow dynamical mode. This distinction is consistent with dissipative topological constructions in which a bulk purity gap closes without a bulk damping-gap closing [7]. Our finite-size hard-wall data exhibit a small stationary occupation gap at $\alpha_1 = 1$ alongside a large channel gap; establishing an actual thermodynamic purity-gap closing requires a separate finite-size analysis.

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